

Project Proposal

Integrated Agricultural Marketplace and Logistics Management System

Project - ICT 2212

Bachelor of Information and Communication Technology

(BICT)

Degree Programme

Department of Information and Communication Technology

Faculty of Technology

Rajarata University of Sri Lanka

Mihintale

Details of the Project

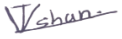
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1. Introduction

Agriculture is the main livelihood for many people in the North Central Province (Rajarata). Even though our farmers produce a large harvest, they face many problems when trying to sell their crops. We noticed that there is a big gap between the farmers in villages and the buyers in big cities. Because of this gap, farmers often don't get a fair price, and a lot of vegetables go to waste before they even reach the market.[1]

Our project, **"HarvestHub,"** is a software solution designed to solve this problem. It is a digital platform that connects farmers directly with buyers Instead of relying on middlemen, our system creates a "Bidirectional Marketplace" where farmers can list what they have, and buyers can post what they need.

The main motivation for this project is to fix the "Trust Gap." Usually, buyers in places like Colombo are afraid to buy from remote farmers because they cannot check the quality of the vegetables. To fix this, our system uses Agricultural Officers to manually check and approve the quality of the harvest before it is sold. We also included a way to manage transport costs automatically so that farmers don't have to worry about finding vehicles.

2. Problem Statement

Our team has analyzed the current agricultural system in the North Central Province (Rajarata) and identified three main problems that our software, **HarvestHub**, will address:

2.1. Information Asymmetry (Farmers Don't Know Demand)

Currently, farmers harvest their crops without knowing the real demand in the market. They often grow the same crops as everyone else at the same time, leading to an oversupply. This causes prices to crash and results in farmers dumping their vegetables because they cannot find buyers.[2]

2.2. The Trust Gap (Buyers Can't Verify Quality)

Buyers in distant cities (like Colombo) are afraid to buy directly from small scale farmers because they cannot verify the quality of the vegetables without seeing them. Because there is no one to guarantee the quality, buyers prefer to use middlemen, which reduces the profit for the farmer.

2.3. Logistical Inefficiency (Transport is Hard to Find)

For small scale farmers, finding a vehicle to transport a small harvest is very difficult and expensive. Often, there is no standard price for transport, and the delay in finding a vehicle causes the fresh produce to spoil before it reaches the market.[3]

2.4. Financial Risk & Payment Uncertainty

In the current system, there is no secure way to handle payments. Farmers are afraid to send goods without money, and buyers are afraid to pay without receiving goods. There is no central system to manage advance payments or ensure fair transactions between strangers.

3. Proposed Solution

HarvestHub is a complete digital ecosystem designed to solve traditional agricultural supply chain inefficiencies. It connects four key users in one unified platform: Farmers, Buyers, Agricultural Officers, and Transporters.

Novelty Compared to Existing Marketplaces

Unlike existing agricultural platforms that only act as simple online classifieds or notice boards (leaving users to independently figure out quality trust, transport, and delivery), HarvestHub introduces three key features:

- **Built-In Logistics Coordination:** Transport is automatically assigned according to harvest capacity, rather than forcing buyers to hire independent trucks.
- **Verified Quality Assurance:** Instead of unverified seller claims, mandatory visual evidence (photos/videos) must be certified by official Department of Agriculture officers before a listing goes live.
- **End-to-End Supply Chain Tracking:** Real-time order and transport tracking from the farm gate directly to the buyer's doorstep.

3.1. For Farmers

Farmers download the mobile app, register with their personal details and exact farm GPS location, and create listings for their harvest. The app guides them to capture clear photos and a short video of their freshly harvested crops. Once submitted, the listing undergoes official quality verification before going live on the marketplace. Farmers can also browse "Wanted" ads from buyers and respond to direct demand.

3.2. For Buyers

Buyers (such as shop owners, supermarkets, or wholesalers) use the app to search for available crops. They can filter listings by produce type, verified quality grade (A, B, or C), They can also post specific supply requests (e.g., *"Need 200kg of Grade A tomatoes by Friday"*). Once a buyer selects a listing, they can negotiate via in-app chat, finalize the purchase, and immediately arrange integrated transport.

3.3. For Agricultural Officers

Officers from the Department of Agriculture access a secure desktop dashboard to review pending crop submissions from local farmers. Only approved and graded listings are published on the marketplace to guarantee buyer trust.

3.3.1. Quality Review Method

To ensure fair and standardized grading without needing to travel to every farm immediately, officers follow a systematic, digital-first evaluation process:

1. **Media Authenticity Check:** The system verifies the timestamp and GPS geotag of the uploaded photos and videos to ensure the media was captured recently at the registered farm location.
2. **Visual Inspection:** Officers examine the images and video footage to assess key visual parameters: **color uniformity**, **size consistency**, **surface blemishes**, and **signs of pest or fungal damage**.
3. **Standardized Grading:** Based on Department of Agriculture guidelines, the officer assigns a grade:
    - **Grade A (Export/Supermarket Quality):** Uniform size and color, zero visible damage or decay, optimal freshness.
    - **Grade B (Standard Retail Quality):** Slight size variation or minor superficial skin blemishes permitted; no rot or structural damage.
    - **Grade C (Processing/Bulk Quality):** Safe for consumption and food processing (e.g., sauces, jams) but lacks visual uniformity or has noticeable aesthetic defects.

3.4. For Transporters

Vehicle owners and independent drivers register on the app to view available delivery jobs. They can accept requests that align with their geographic route and vehicle weight capacity. The system automatically calculates a fair, transparent delivery fee based on exact travel distance and cargo weight, reducing transit times and post-harvest spoilage.

4. Scope of the project

The HarvestHub project is a digital solution designed to fix the current gaps in the agricultural supply chain in the North Central Province. The scope covers building a "Trust Ecosystem" that connects farmers, buyers, and transporters through a verified and managed platform.

4.1. In-Scope (Main Deliverables)

These are the specific features and activities the team will complete during the project:

Table 1: Main deliverables of Project (in scope)

Feature	Description
User Registration & Roles	We will create role-based access for four distinct users: Farmers, Buyers, Transporters, and Agricultural Officers.
Bidirectional Marketplace	This includes a search and filter system where Farmers list their harvest (Supply) and Buyers post "Wanted" ads (Demand).
Manual Quality Verification	A JavaFX desktop portal for Agricultural Officers to manually review photos and approve crop quality grades (e.g., Grade A).
Automated Vehicle Selector & Logistics Calculator	An algorithm that automatically assigns the most suitable transport vehicle and calculates dynamic transport fees based on the total harvest weight and the travel distance.
Order Workflow Engine	Logic to handle the full transaction cycle, including price agreement, advance payments, and final delivery clearance.
2-Way Digital Handshake	A security feature where an order is marked 'Complete,' and funds are automatically released to the Farmer only after synchronized delivery confirmation from both the **Transporter** and the **Buyer** at the physical drop-off location."
Trust-Based Rating System	A reputation module where Farmers and Buyers rate each other based on their transaction history.
Live Tracking	Integration of Google Maps API to allow Buyers to track their shipments in real-time.

4.2. Excluded from Scope (Project Boundaries)

Table 2: Project Boundaries (out of scope)

Feature	Description
AI Quality Check	We will not use AI or image processing to automatically grade crops; this is deferred for future implementation
Live Payment Gateway	Real-world bank integration is excluded; we will use mock (simulated) payments for demonstration purposes.

5. Aim and Objectives

5.1. Aim

The aim of this project is to develop a trusted agricultural marketplace and logistics ecosystem that connects farmers directly with buyers, ensuring fair trade through officer-verified quality checks, automated transport cost calculation, and a secure payment-first transaction workflow.

5.2. Objectives

The project will be achieved through the following six specific and measurable objectives:

1. To develop a GPS-Verified Crop Listing Module: To build an interface for farmers to view current market demand and create crop supply listings using real-time, location-tagged photos to ensure authenticity.

2. To implement an Officer Verification Dashboard: To build a centralized desktop portal for Agricultural Officers to manually review, grade (e.g., Grade A), and approve crop photos and GPS data before they are published to buyers.

3. To create a Marketplace Negotiation System: To develop an application module that enables direct communication between farmers and buyers to negotiate prices and securely schedule pickup dates.

4. To build an Automated Logistics and Dispatch Engine: To implement a system that automatically calculates transport costs based on weight and distance, assigns suitable vehicles, processes advance payments, and dispatches drivers.

5. To build a Pre-Loading Quality Verification Feature: To create a checkpoint in the Transporter App where the transport helper inspects the crops and uploads photos, allowing the buyer to remotely approve or reject the harvest quality before the advance payment is released to the farmer.

6. To implement a 2-Way Final Delivery Confirmation: To build a final step where only the transporter (confirming drop-off) and the buyer (confirming receipt) verify the delivery, automatically completing the order and releasing final payments without requiring any manual confirmation from the farmer.

6. Technical Approach

6.1. Overall System Architecture

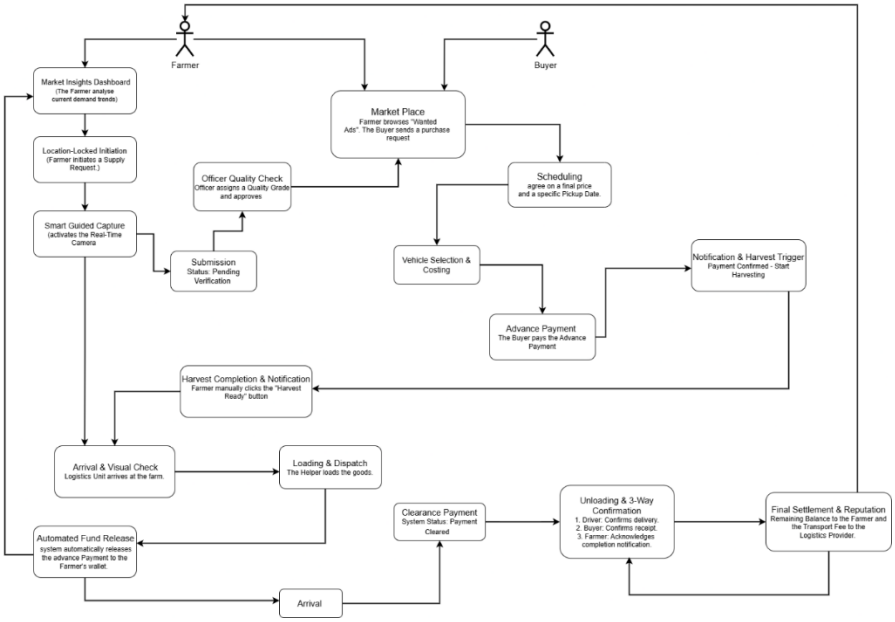


Figure 1: System Workflow

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6.2. Layered Architecture of the system

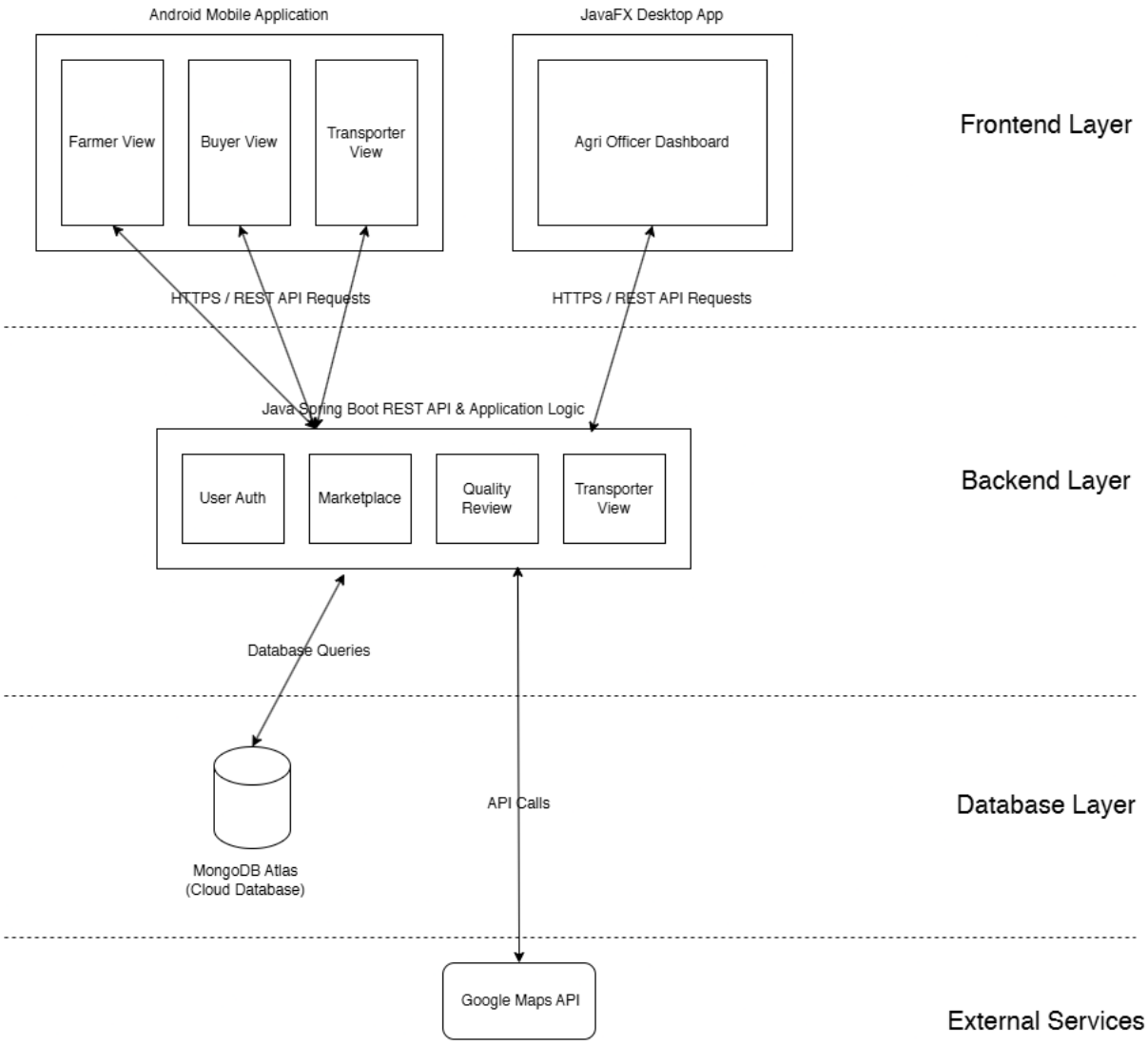


Figure 2: Layered Architecture of the system

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The system will adopt a multi-tier architecture with separate front-end, back-end, and database layers to ensure scalability, security, and maintainability.

Architecture Layers:

1. **Frontend Layer**
   - JavaFX Desktop Application (Admin & Agri officer)
   - Android Mobile Application (Farmers, Buyers, Transporters)
2. **Backend Layer**

- REST API developed using Java Spring Boot

3. Database Layer

- MongoDB Atlas (Cloud Database)

4. External Services

- Google Maps API (Distance & Route Calculation)

6.3. Frontend Design & Technologies

6.3.1.JavaFX (Admins)

- Used by Admin and Agri officers and for:
    - Manual quality verification of crop listings
    - Approving or rejecting listings
    - Monitoring users, orders, and disputes
- Features:
    - Secure login with role-based access
    - Dashboard with tables and filters
    - Listing approval workflow
- Communication:
    - JavaFX client communicates with backend using REST APIs

6.3.2.Android Mobile Application (Java + XML)

- Used by:
    - **Farmers**: Upload crop listings
    - **Buyers**: Post demand requests & place orders
    - **Transporters**: View delivery tasks
- Features:
    - Role based UI
    - Crop listing & buyer request creation
    - Order tracking and confirmation
    - Rating and feedback submission

6.4. Backend Design (Spring Boot)

6.4.1.Core Backend Responsibilities

- User authentication & authorization
- Marketplace logic
- Order workflow management

- Logistics cost calculation
- Rating and trust score generation

6.4.2.Security

- Spring Security with JWT-based authentication
- Role-based access control:
  - Farmer
  - Buyer
  - Transport
  - Admin

6.5. Database Design

The system utilizes a cloud-based NoSQL document database to ensure scalability, real-time synchronization, and flexible data structures. The data is organized into the following primary collections:

- **Users Collection:** Stores user documents including unique ObjectId, user role (Farmer, Buyer, Driver, Officer), contact information, and dynamically updated trust rating scores used to evaluate reliability.
- **Listings Collection:** Stores farmer crop submissions as documents. This includes the unique ObjectId, a reference to the farmer, embedded geospatial data (GPS coordinates), media links for the Smart Camera photos, available quantity, price, and the current verification status (e.g., *Pending, Under Review, Approved*).
- **Market Requests Collection:** Stores buyer demand documents, containing the buyer reference, required crop type, requested quantity, and target location.
- **Orders Collection:** Acts as the central transaction record. It stores the current state-machine status (e.g., Awaiting Payment, In Transit, Completed), advance payment confirmations, and 2-way completion signatures.
- **Logistics & Tracking Collection:** Stores transportation metadata linked to specific orders, including calculated travel distance in kilometers, total transport cost, and embedded location data for route tracking.
- **Chat Collection:** Efficiently stores real-time messaging documents between buyers and farmers for negotiation and pickup scheduling.

6.6. Bidirectional Marketplace Logic

- Farmers post Supply Listings
- Buyers post Demand Requests
- Search and filtering based on:

- Crop type
- Location
- Quantity
- Price range

- Matching handled at backend using SQL queries and filters

6.7. Collaborative Quality Verification Queue

- Implemented using JavaFX · Workflow:
- Farmer submits crop listing
- Status = **PENDING** (Visible in a global queue to all Agricultural Officers)
- Available Officer **claims** the listing from the queue
- Status changes to **UNDER_REVIEW** (Listing is temporarily locked to this officer to prevent duplicate grading)
- Officer reviews the GPS data and Smart Camera photos
- Approves (assigns Quality Grade), Rejects, or Requests a Photo Retake · Approved listings become visible in the marketplace to buyers

6.8. Algorithmic Vehicle Selector & Dynamic Cost Calculator

- Uses Google Maps API
- Fetch farmer and buyer GPS coordinates
- Calculate total travel distance in Km via Google Maps API
- Analyze total harvest weight to automatically assign the most suitable vehicle type (e.g., matching a 1-ton lorry to an 800kg harvest)
- Calculate final dynamic transport fee using the multi-factor pricing formula (Distance + Weight + Fuel rate) · Final transport details are stored as a document in the Logistics & Tracking Collection (MongoDB)

6.9. Advanced Order Workflow & Liability Engine

The order lifecycle is managed using strict backend state-machine transitions to ensure financial security and accountability:

- **Automated Dispatch & Escrow Payment** (Locks funds and loops for available drivers)
- **Proof-of-Loading Checkpoint** (Establishes liability baseline before transit)
- **In-Transit Telemetry & Lock** (Continuous GPS tracking)

- **2-Way Digital Handshake**
    - o Transporter (Confirms GPS drop-off)
    - o Buyer (Confirms visual quality)
- **Multi-Stage Dispute & Escrow Reversal** (Handles Pre-loading rejections, Post-delivery rejections, and Reverse Logistics)

Order Status Values (Backend State Nodes):

- **Pending** (Awaiting Buyer payment)
- **Paused** (Polling for available transport vehicles)
- **Loaded & Verified** (Proof-of-loading evidence accepted)
- **In Transit** (Order locked; live telemetry active)
- **Delivered** (Awaiting 2-Way Handshake clearance)
- **Completed** (Escrow automatically disbursed to wallets)
- **Rejected / Disputed** (Escrow frozen; automated refunds triggered)
- **Return Journey** (Reverse logistics protocol active)

6.10. Rating & Trust Score System

- Users rate each other after order completion
- Backend calculates:
    - Average rating
    - Transaction count
    - Completion reliability
- Trust score displayed in user profiles

6.11. Project Methodology – Incremental

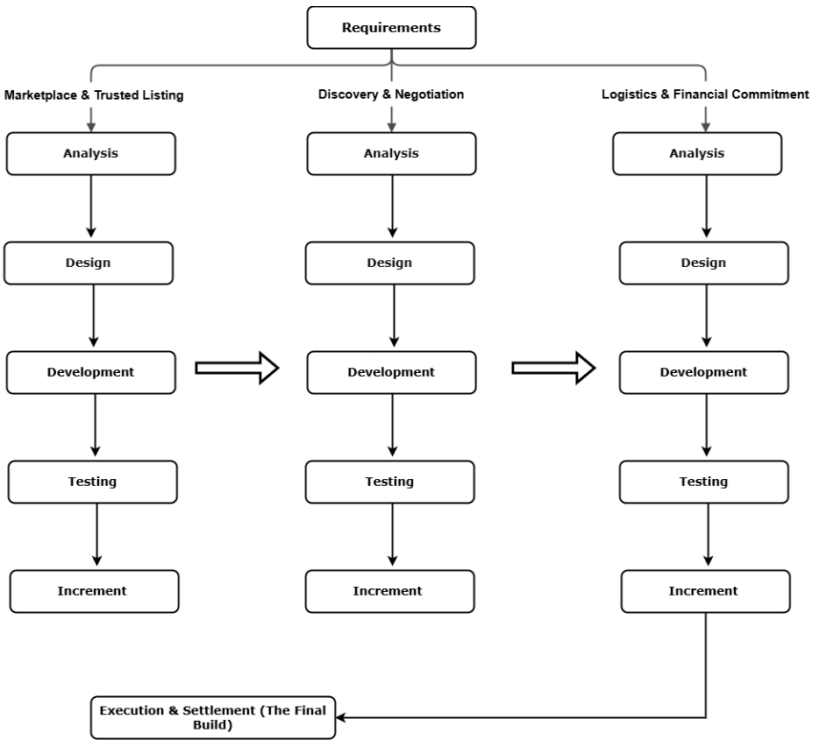


Figure 3:Project Methodology (Incremental Model)

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We have chosen the incremental model for our project. Below are some of the main reasons for choosing it.

1. **Early Delivery of the Marketplace**

   While we are still developing complex logistics and payment algorithms, farmers can start registering their farms and "officially-verifying" their crops. This builds a logistics database early in the project lifecycle.

2. **Complexity Management of Multi-Role Interactions**

   Trying to build all these interfaces at once is very complex. With this incremental model we have:

   - First, stabilize the **Farmer/Officer** relationship (Increment 1).
   - Then, introduce the **Buyer** negotiation (Increment 2).
   - Finally, integrate the **Driver/Helper** workflow (Increment 3)

3. **Better Handling of Physical Verification**

This is a technical challenge. By reserving this for a later increment, we ensure that the "smart guidance capture" in Increment 1 has already been perfected and produces high-quality geo-tagged data for the driver to use during the verification phase.

6.12. Business & Revenue Model

To ensure long-term financial sustainability while driving massive user adoption, HarvestHub employs a "Split Transaction Fee" model. The system acts as an automated accountant, securely processing all payments and strictly deducting service commissions **only upon successful deliveries**. This ensures platform goals are perfectly aligned with user satisfaction.

- **Buyer Convenience Fee (3%):** Applied to the Buyer's total order value at checkout. This fee monetizes the convenience of bypassing traditional middleman markups, saving travel time, and securing verified quality crops.
- **Transporter Commission (5%):** Deducted automatically from the final delivery fee. HarvestHub acts as a guaranteed lead generator for logistics providers, eliminating empty return trips and idle stand time, which easily justifies the commission.
- **Farmer Fee (0%):** To heavily incentivize agricultural sector adoption and rapidly build the platform's supply side, Farmers are charged zero platform fees for listing or selling their harvests.
- **Refund Policy (Zero-Fee on Failure):** If a transaction results in a rejection or dispute, the system automatically waives all platform fees, ensuring the Buyer receives a 100% refund of both their core payment and the convenience fee to maintain absolute trust.

7. Project Work Plan

The Project Work Plan describes how the project will be managed, including a detailed timetable with milestones. To ensure the "HarvestHub" system is built efficiently, we have planned a 12-month schedule using an **Incremental** development approach.[4]

7.1. Project Timeline and Milestones

The following table provides a breakdown of the key tasks, timelines, and the expected outputs for each phase of the project.

Table 3: Project Work Plan

2.0	**Increment 01: Trust & Verification Build**	**Month 3-5**	**Core Verification Module**

2.1	Environment Setup	Month 3 (Week 1)	Project Repo & Server configuration
2.2	Backend: Auth System (JWT) & APIs	Month 3 (Week 2-4)	Secure Login & DB Connectivity
2.3	Android: Farmer View & Smart Camera	Month 4	Guided Capture & GPS Locking
2.4	JavaFX: Admin Panel (Officer View)	Month 5 (Week 1-3)	Desktop interface for crop grading
2.5	Inc. 01 Integration Testing	Month 5 (Week 4)	Testing Farmer Upload & Approval

2.0	**Increment 01: Trust & Verification Build**	**Month 3-5**	**Core Verification Module**
2.1	Environment Setup	Month 3 (Week 1)	Project Repo & Server configuration
2.2	Backend: Auth System (JWT) & APIs	Month 3 (Week 2-4)	Secure Login & DB Connectivity
2.3	Android: Farmer View & Smart Camera	Month 4	Guided Capture & GPS Locking
2.4	JavaFX: Admin Panel (Officer View)	Month 5 (Week 1-3)	Desktop interface for crop grading
2.5	Inc. 01 Integration Testing	Month 5 (Week 4)	Testing Farmer Upload & Approval
3.0	**Increment 02: Marketplace Build**	**Month 6-8**	**Bidirectional Trading**
3.1	Android: Buyer View (Search & Filter)	Month 6	Listings Feed UI & Filtering Logic
3.2	Backend: Chat & Scheduling Logic	Month 7 (Week 1-2)	Socket.io Setup & Pickup Logic
3.3	Android: In-App Chat Module	Month 7 (Week 3-4)	Chat UI for Negotiation
3.4	Backend: Order State Machine (v1)	Month 8 (Week 1-2)	Status: Pending → Accepted
3.5	Inc. 02 System Testing	Month 8 (Week 3-4)	Testing Negotiation Flow
4.0	**Increment 03: Logistics Ecosystem Build**	**Month 9-11**	**Automation & Finance**
4.1	Backend: Logistics Algo	Month 9	Cost Calculation Logic ($/km $/kg)
4.2	Android: Transporter View	Month 10 (Week 1-2)	Driver UI & Status Updates
4.3	Android: Helper Verification Feature	Month 10 (Week 3-4)	Pre-Loading Check Interface
4.4	Backend: Payment & Rating Logic	Month 11 (Week 1-3)	Trust Scores & Fund Release
4.5	Inc. 03 Full Integration Testing	Month 11 (Week 4)	End-to-end workflow testing
5.0	**Phase 3: Finalization & Deployment**	**Month 12**	**Project Closure**
5.1	User Acceptance Testing (UAT)	Month 12 (Week 1-2)	Final bug fixes and optimization
5.2	Final Report / Thesis Writing	Month 12 (Week 3)	Completing documentation
5.3	Viva Presentation Preparation	Month 12 (Week 4)	Preparing slides and demo video

7.2. Gantt Chart Description

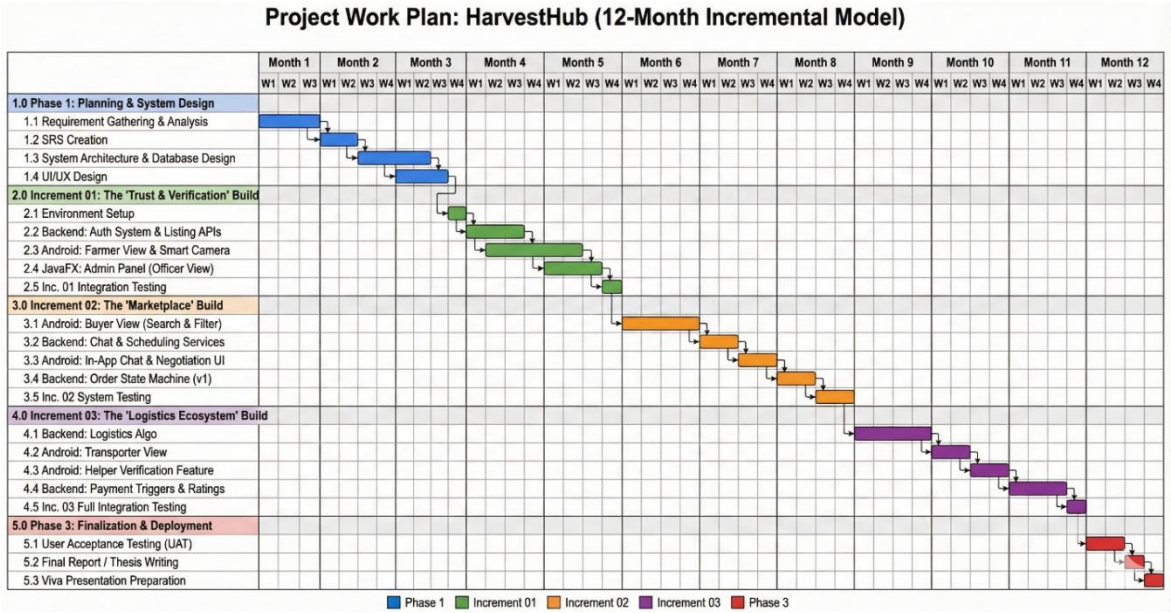


Figure 4: Gantt Chart of the project work plan

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8. Project Resources

8.1. Human Resources

The project is executed by a team of six undergraduates, covering the following key roles:

- **UI/UX Designers:** Responsible for designing user-friendly interfaces and high-fidelity prototypes.
- **Development Team:** Android (Mobile), Spring Boot (Backend), and JavaFX (Desktop) development.
- **Quality Assurance (QA) Engineers:** Responsible for system testing, specifically verifying GPS accuracy and camera modules.
- **Project Supervisor:** Provides academic guidance and monitors project milestones.

8.2. Physical & Software Resources

8.2.1.Hardware

Development laptops and physical Android smartphones are required to test hardware-dependent features, such as **real-time GPS tracking and camera-based verification**.

8.2.2.Software Stack:

- **Development:** Android Studio, IntelliJ IDEA, MongoDB Compass
- **APIs & Testing:** Google Maps API (Location Services), Postman (API Testing).
- **Design & Collaboration:** Figma (UI Design), Git/GitHub (Version Control).

9. Conclusion

In conclusion, the "HarvestHub" project is designed to provide a practical solution to the real-world problems faced by the agricultural community in the North Central Province. By connecting farmers, buyers, transporters, and agricultural officers through a single digital platform, we aim to eliminate the middleman and build a transparent "trust ecosystem".

The system focuses on solving the major issue of crop quality through a manual verification portal. It also simplifies logistics by automating transport cost calculations based on weight and distance. By following a structured 12-month work plan and using modern technologies like Spring Boot and Android, our team is confident that we can deliver a functional system that benefits boththe industry and our technical skills development.

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